ENPM 701 – Autonomous Robotics: Spring 2026

Master of Engineering Program in Robotics





Autonomous agricultural technology ("ag tech" for short) uses robotic harvesters, AI-powered tractors, and precision localization systems to automate a wide range of daily activities, from planting, to monitoring, to harvesting, and more. These technologies help farmers increase yields while reducing labor and input costs, leading some industry reports to estimate the market to be on the order of $75 billion this year alone.

Companies like Burro (top, left) Yarbo (top, right) and Greenfield Robotics (bottom) are leading the shift toward data-driven, self-operating farm systems that make agriculture more efficient, profitable, and safer. Combined with advances in electric vehicle technologies and wireless charge capabilities, sustainabilty is leading the way in ag tech.



Studying the autonomous ag tech industry puts us at the forefront of one of the world's most important industries - where innovation directly impacts food security, climate resilience, and global economic stability. In this spirit, the Grand Challenge (GC) for ENPM701 involves the design and implemenation of a wheeled robot for agriculatural activities.

Students in ENPM701 will design and implement a four-wheeled ground robot that traverses a simulated agricultural center. The robot will assist farmworkers with repetitive, labor-intensive tasks such as harvest transport, bin and crate movement, and supply deliveries.

GC Learning Outcomes

- Apply kinematic models to design and control a mechanical wheeled robot
- Apply control system theory to design and implement a robot control system
- Integrate sensors and actuators into a mechatronic system
- Implement intelligent robotics algorithms, including image processing and motion control and planning

What the GC Involves

- Navigation – traverse autonomously through a cluttered environment in order to find objects and transport them to a specified location. Using known coordinates of landmarks, the robot must use localization techniques to plan the best course to traverse.
- Spatial Awareness – the robot must be aware of its surroundings in order to avoid unwanted situations. It must be able to detect, differentiate, and avoid/interact with objects.
- Efficiency – be able to complete the given task in the most efficient way possible, particularly within the given temporal constraints.
- Strategy – employ an effective strategy to optimize speed, task planning, etc.

Grand Challenge Rules & Specifications

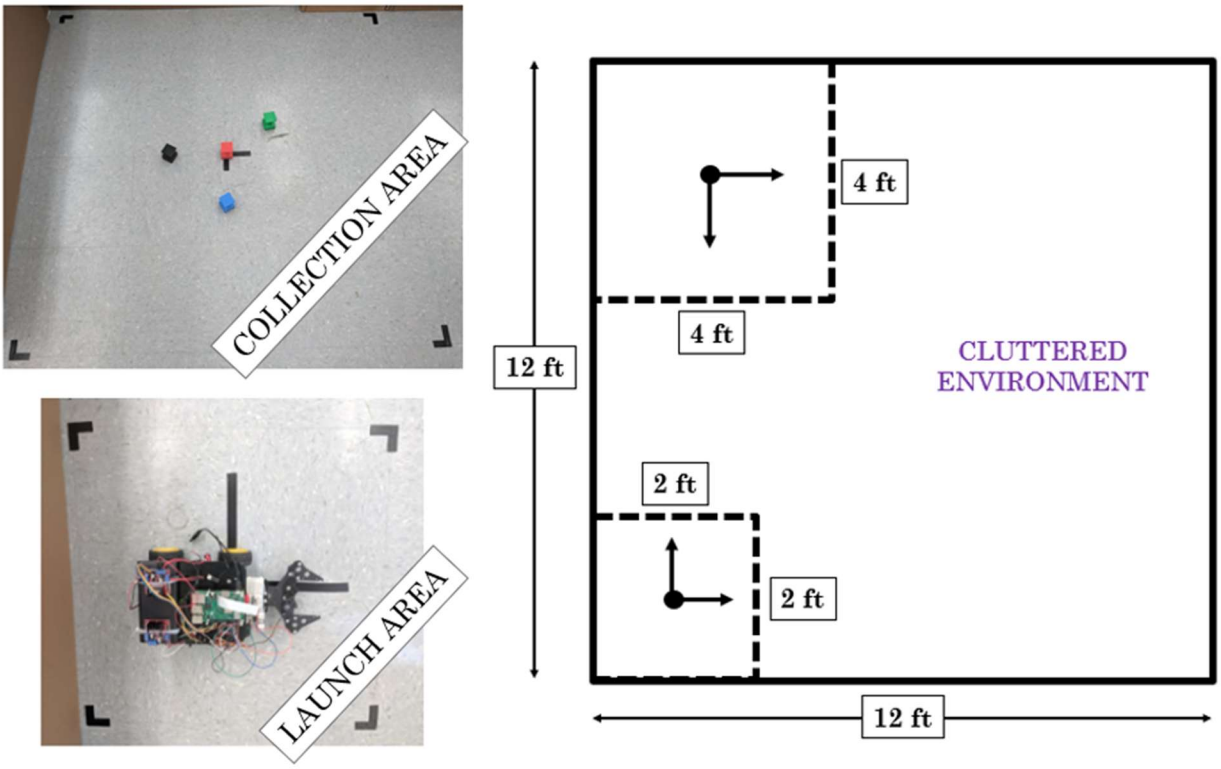
Robot

- The base mobile platform will be the same for each student and is available for purchase from the OAEE office. The mobile platform kit includes the frame, motors, wheels, motor drivers, Raspberry Pi + camera, sonar range sensors, and servo gripper assembly. Students must acquire/install all other required/desired components as well as write code to control the robot.
- Only one robot may perform at a time.
- Each student must construct their own robot.
- Each robot must have a safety shutdown switch.
- When needed, students must communicate with their robot wirelessly. Wi-Fi is the preferred wireless communication method. Participants may set up their own Wi-Fi

networks (e.g. cell phone hotspot) or connect through UMD Wi-Fi.

Testing area

- Exact dimensions of the testing area are shown below. Note: these dimensions may change based on laboratory availability. Dr. Mitchell will confirm the agreed-upon final dimensions in advance of the Grand Challenge.
- To start a run through the Challenge arena, the vehicle will be placed in the center of the launch area, at a heading of the user's choosing:



- Walls are made of cardboard or equivalent and colored black.
- All obstacles are fixed and immovable.
- Floor of the testing area is flat and even and colored white/beige (tile).

Blocks

- Each block is a basic cubic shape with a vertical gripping arm. Solid model files of the blocks are available under ELMS > Project. The geometry of each block is the same.
- Each block can be either Red, Green, or Blue, but no combination of more than one color. It is also possible for blocks to be located in the Cluttered Environment of White, Black, or Yellow color.
- Each student will be provided with a single block of Red, Green, and Blue during the course to use for testing their algorithms in advance of the GC
- Blocks must be delivered by the robot to the predetermined collection zone.

Points

- Students will receive a single point for each Red, Green, or Blue block delivered inside the collection zone. The entirety of the block must be within the collection zone bounds once the block comes to rest.
- Students will be deducted a single point for each White, Black, or Yellow block delivered inside the collection zone, in part or in full.

Procedure

- Students have full access to their robots prior to the beginning of each run, at which point they may only receive data during the run
- Each robot starts the run in the launch zone. Dr. Mitchell will email each student's robot to initiate the run, at which point the clock starts (Dr. Mitchell will keep track of the clock).
- During the run robots must explore the cluttered area, find blocks and deliver them to the collection zone.
- Each run ends if any of the following occurs:
    - robot malfunctions;
    - the student calls to stop the run;
    - all blocks were located and delivered;
    - 10 minutes elapsed.
- Students have 2 attempts to complete a run. The run with highest number of blocks transmitted to the collection zone will be used for scoring.
- Students cannot interfere in any way with other students' runs.

- During each run, the student must be in the control room and reference only the data transmitted from the robot (…remember, the robot is on Mars!).
- If a robot breaks through the walls or leaves the testing area during the run, the run ends immediately.
- Students cannot set up any additional external devices (beacons, cameras, etc) to aid the run, outside of those provided.

General

- Students may collaborate during development of their robots, however each student must **build <u>their own</u> robot** and **develop <u>their own</u> codes**
- Students are free to acquire additional sensors/actuators to add to their robot throughout the semester

Deliverables

- **Due: Friday 5/15/26** via email submission to Dr. Mitchell, NLT **11:59pm** local time. Include in the email:
    - a link to your YouTube video
    - an attached file(s) detailing all codes used during the Grand Challenge

Video Details

A key component of the engineering practice, admittedly often insufficiently addressed in academic curricula, is the link between technical activities and communications/marketing. Often the most successful engineers are those who not only design and deliver cutting edge technologies, but perhaps more importantly effectively communicate technical ideas and relevant financial/societal impacts to a larger audience.

In this spirit, each student is required to create and upload a video to your individual YouTube account and provide the link to it (the video can remain unlisted, which means only people with the link can see it, if you wish). Your video (in its entirety, or a portion) may be showcased in future OAEE publications and presentations. So, spend some time on the video and show off your work! **If nothing else, have fun with the filming process!**

The video must be no less than eight (8) minutes and no longer than ten (10) minutes in length and cover the following:

1. Overall
    a. Minimum eight (8) minutes, maximum ten (10) minutes in length
    b. Visual and audio clarity
    c. Creativity
2. Introduction
    a. Background of the course and the Grand Challenge.
3. Robot Design
    a. Description of robot hardware
    b. Description of robot software
    c. This section should demonstrate your understanding of the fundamentals of autonomous robotics, as applied to your vehicle's hardware/software and the Grand Challenge
    d. Vehicle integration/test/calibration activities
    e. Description of any challenges encountered (e.g. technical, hardware/software obstacles, etc.) and how you overcame those challenges to complete the GC
4. Grand Challenge
    a. Performance of your robot during the GC
    b. Time-lapse of GC performance from robot's perspective
    c. Data/results pertinent to the GC in the context of the course, such as map of block locations

Suggestion: use your phone to record 30-60 second video clips throughout the semester, then stitch them together towards the end of the semester. *General theme of engineering documentation*: take images and video all the time, as you go, then edit as desired later on. **Always better to cut out extra material than to say "I wish I had an image of that…!" at the conclusion of the project**.

Examples of exemplary project student videos from other/related courses:
https://www.youtube.com/watch?v=I1i44fPISbw&list=PLekqOMAzgrTRP6Ms7wI37-2lKjU92tqnL

https://www.youtube.com/watch?v=AiWkhdPCsyo&list=PLekqOMAzgrTQJ5O1nEiZj_Gur7ZQvq-4N

*Please watch immediately: **Vertical Video Syndrome - A PSA**
https://www.youtube.com/watch?v=dechvhb0Meo